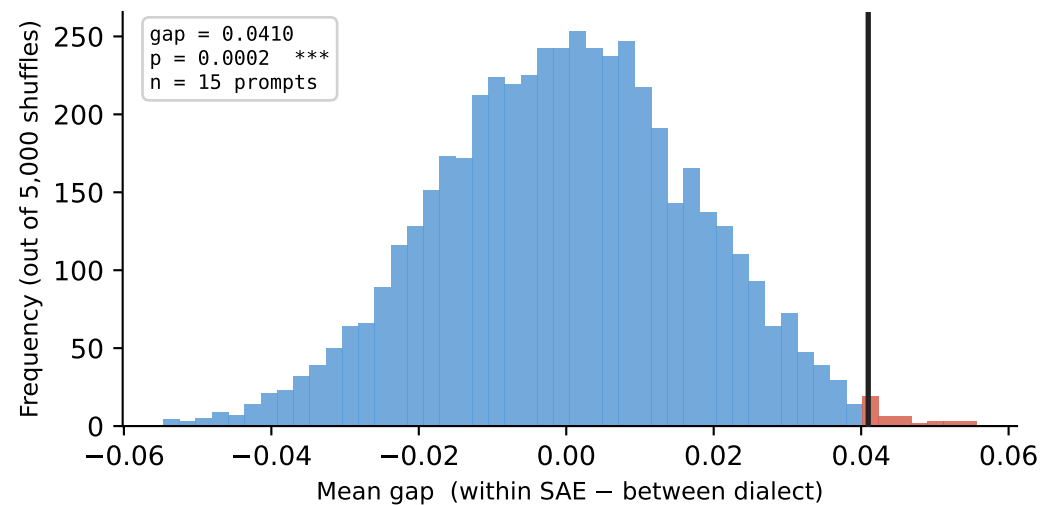
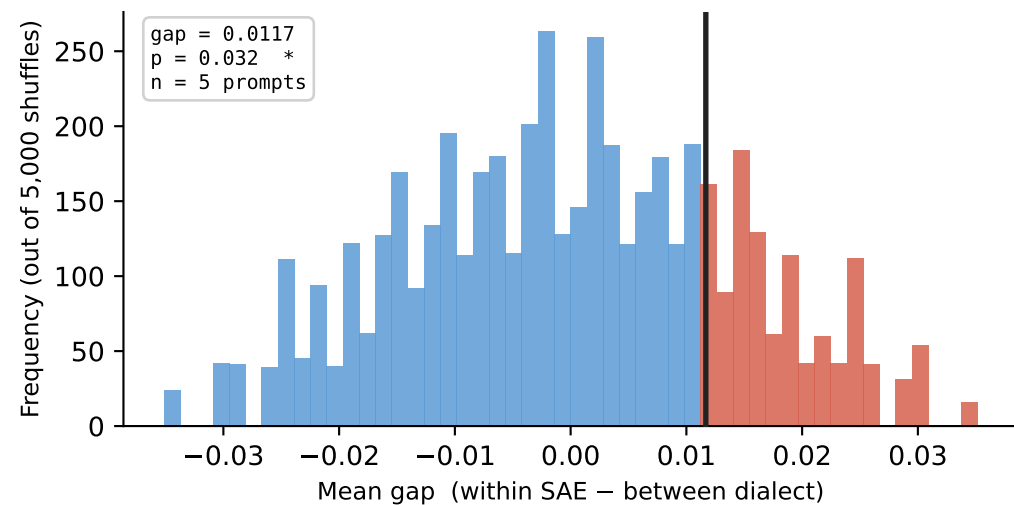


**Permutation Test Null Distributions — Within-SAE vs Between-Dialect Gap
(5,000 shuffles per test; reported p-values from original pair-level test)**

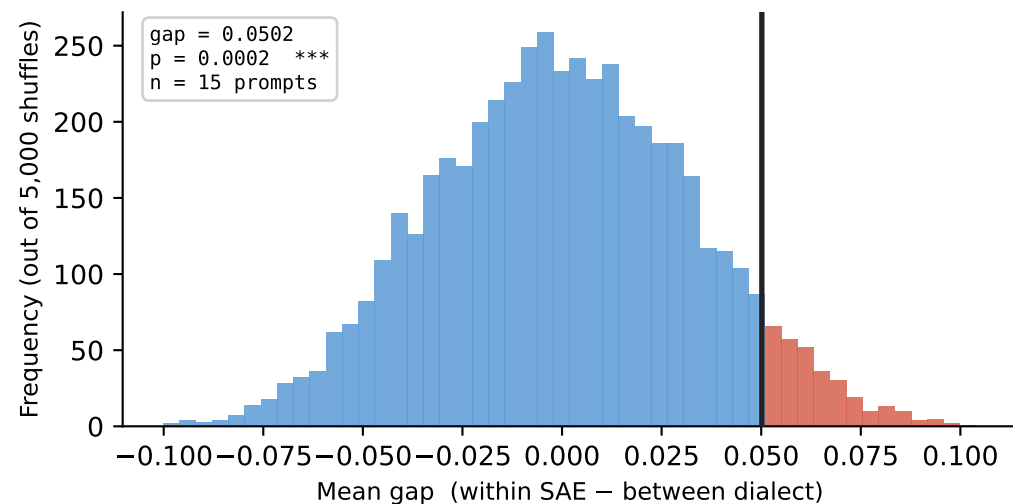
Round 1 — Cosine



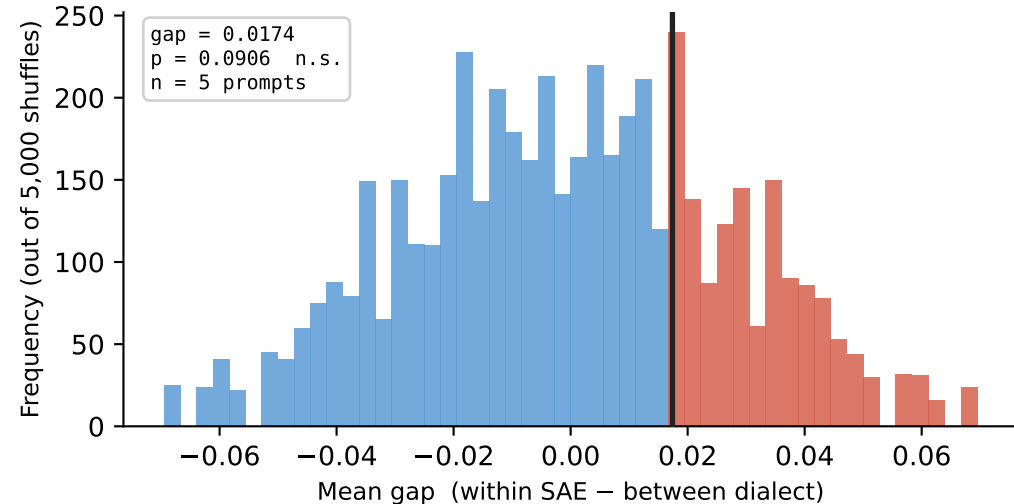
Round 2 — Cosine



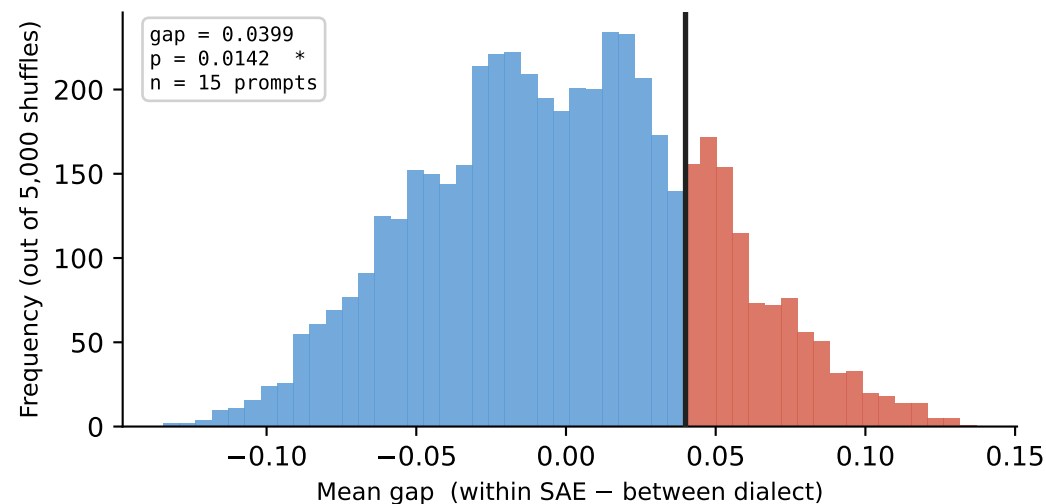
Round 1 — Token J



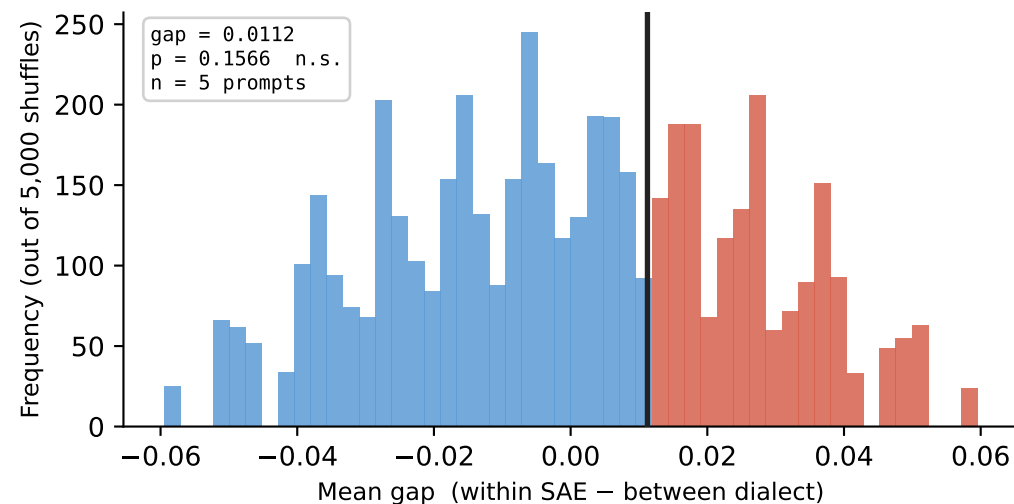
Round 2 — Token J



Round 1 — Bigram J



Round 2 — Bigram J



Null distribution (shuffled gaps) Tail (≥ observed gap; area = p-value) Observed gap (real within-SAE – between-dialect)